EEN 133 HW 4

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1. i + ii

	L = 5	25	125
# of 0s	5-1 = 4	25-1=24	125-1=124
first zero	$\frac{2\pi}{5}$	$\frac{2\pi}{25}$	$\frac{2\pi}{125}$

iii.

The first zeros are found from $w = \frac{2\pi}{L}$

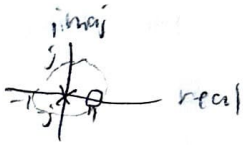
(for all graphs)

2.

a. $x[n] = \delta(n) - \delta(n-1) \rightarrow X(z) = 1 - z^{-1}$

$z = 1 \rightarrow$ zero

$z = 0 \rightarrow$ pole



b. $x(n) = \delta(n) - \delta(n-2) \rightarrow X(z) = 1 - z^{-2}$

$z = 1, -1 \rightarrow$ zero

$z = 0 \rightarrow$ pole

x2

same pole, -1, 1 are zeros now

c. $x(n) = \delta(n) + \delta(n-2) \rightarrow X(z) = 1 + z^{-2}$

$z = j, -j$ zero

$z = 0 \rightarrow$ pole

x2

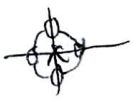
same pole, imaj imag zeros.

d. $x(n) = \delta(n) - \delta(n-4) \rightarrow X(z) = 1 - z^{-4}$

$z = j, -j, 1, -1 \rightarrow$ zeros

$z = 0 \rightarrow$ pole

x4



has all of b and c's zeros and poles

b's $x(n)$ * c's $x(n)$ = d's $x(n)$

e. $x(n) = \delta(n) + \delta(n-4) \rightarrow X(z) = 1 + z^{-4}$

$1 + z^{-4} = 0$

$z^{-4} = -1$

$z = \pm\frac{\sqrt{2}}{2} \pm \frac{\sqrt{2}}{2}j$

$z = \pm\frac{\sqrt{2}}{2} \pm \frac{\sqrt{2}}{2}j \rightarrow$ zeros

$0 \rightarrow$ pole

x4

This result is part d but rotated essentially $\pi/4$ rad.

3.

a. $y(n) = x(n) + 2x(n-1) \rightarrow H(z) = 1 + 2z^{-1}$

poles for all at $z=0$

zeros: $z=-2$

b. $y(n) = y(n) + .81x(n-2) \rightarrow H(z) = 1 + .81z^{-2}$

$z = \frac{\sqrt{-.81}}{1} = \pm .9j$

$z = \pm j\frac{9}{10}$

c. $y(n) = x(n) + x(n-1) + x(n-2) \rightarrow H(z) = 1 + z^{-1} + z^{-2}$

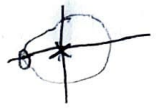
$1 + \frac{1}{z} + \frac{1}{z^2} = 0$ $\quad z^2 + z + 1 = 0$

$\frac{-1 \pm \sqrt{1-4}}{2}$

$z = \frac{-1 \pm j\sqrt{3}}{2}$

d. $y(n) = x(n) + 2x(n-1) + x(n-2) \rightarrow H(z) = 1 + 2z^{-1} + z^{-2}$

$z = -1$



e. $y(n) = y(n) + 5x(n-1) + x(n-2) \rightarrow H(z) = 1 + 5z^{-1} + z^{-2}$

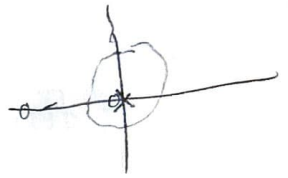
$1 + \frac{5}{z} + \frac{1}{z^2} = 0$

$z^2 + 5z + 1 = 0$

$\frac{-5 \pm \sqrt{25-4}}{2}$

$-\frac{5}{2} \pm \frac{\sqrt{21}}{2}$

$z = -\frac{5}{2} \pm \frac{\sqrt{21}}{2}$



4.

a. $y(n) = x(n) + .5y(n-1)$

$Y(z) = X(z) + .5z^{-1}Y(z)$

$Y(z)(1 - .5z^{-1}) = X(z)$

$H(z) = \frac{1}{1-.5z^{-1}} = \frac{z}{z-.5}$

zero: $z=0$

pole at $z=.5$

stable b/c $|.5| < 1$

$\frac{1}{1-.5e^{jw}}$ $\quad e^{jw} = \cos w - j\sin w$

$\left|\frac{1}{1-.5(\cos w - j\sin w)}\right|$ only care about real part

$\frac{1}{1-.5\cos w}$

max 2 at $w=0$

min $\frac{2}{3}$ at $w=\pi$

b. $y(n) = x(n) - x(n-1) + 2y(n-1)$

$Y(z) = X(z) - z^{-1}X(z) + 2z^{-1}Y(z)$

$Y(z)(1-2z^{-1}) = X(z)(1-z^{-1})$

$H(z) = \frac{1-z^{-1}}{1-2z^{-1}} = \frac{z-1}{z-2}$

0 at $z=1$

pole at $z=2$

unstable b/c $|z| \geq 1$

Not quite sure what the bounded input sequence means, but I'll guess $x(n) = u(n)$

c. $y(n) = x(n) + y(n-1)$

$H(z) = \frac{z}{z-1}$

0 at $z=0$

pole at $z=1$

unstable b/c $|z| \geq 1$.

example input: $x(n) = 2u(n)$?

d. $y(n) = x(n) - y(n-1)$

$Y(z) = X(z) - z^{-1}Y(z)$

$Y(z)(1+z^{-1}) = X(z)$

$\frac{1}{1+z^{-1}} = \frac{z}{z+1}$

0 at $z=0$

pole at $z=-1$

unstable b/c $|z| \geq 1$

Example input: $x(n) = 2u(n)$

e. $y(n) = \frac{1}{2}x(n) - \frac{1}{2}(x(n-2)) + \frac{81}{100}y(n-2)$

$Y(z) = \frac{1}{2}X(z) - \frac{1}{2}z^{-2}X(z) + \frac{81}{100}z^{-2}Y(z)$

$Y(z)(1 - \frac{81}{100}z^{-2}) = X(z)(\frac{1}{2} - \frac{1}{2}z^{-2})$

$H(z) = \frac{.5 - .5z^{-2}}{1 - .81z^{-2}} = \frac{.5z^2 - .5}{z^2 - .81}$

0s at $z = \pm 1$

poles at $z = \pm .9$

stable b/c $|z| < 1$ at poles

$\frac{.5e^{2jw} - .5}{e^{2jw} - .81} = \frac{.5(\cos 2w + j\sin 2w) - 1}{\cos 2w + j\sin 2w - .81}$

$\frac{.5(\cos 2w - 1)}{\cos 2w - .81}$

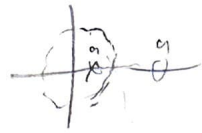
$w = 0, \pi$, min $= 0$

$w_1 = \frac{\pi}{2}, \frac{3\pi}{2}$, max $= .553$

5. $H(z) = \dfrac{a - z^{-1}}{1 - a z^{-1}}$ $\quad 0 < a < 1$

$= \dfrac{az - 1}{z - a}$

a.



pole at a

zero at $\frac{1}{a}$

b. $Y(z)(1 - a z^{-1}) = X(z)(1 - z^{-1})$

$Y(z) - a z^{-1} Y(z) = a X(z) - X(z) z^{-1}$

$y(n) = a x(n) - x(n-1) + a y(n-1)$

c. $H(e^{jw}) = \dfrac{a - e^{-jw}}{1 - a e^{-jw}}$

d. $|a - e^{-jw}| = |a - (\cos w - j \sin w)| = |a - \cos w + j \sin w| = \sqrt{(a - \cos w)^2 + \sin^2 w}$

phase:

real: $a - \cos w$

imaj: $\sin w$

$\boxed{\text{phase: } \tan^{-1}\left(\dfrac{\sin w}{a - \cos w}\right)}$

$\sqrt{a^2 - 2a\cos w + \cos^2 w + \sin^2 w} = \sqrt{a^2 - 2a\cos w + 1}$ ← mag

e. $|1 - a e^{-jw}| = |1 - a(\cos w - j \sin w)| = |1 - a\cos w + a j \sin w|$

phase:

real: $1 - a\cos w$

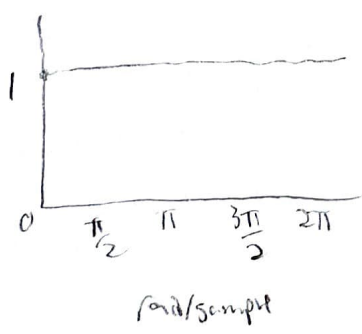
imaj: $a \sin w$

$\boxed{\text{phase: } \tan^{-1}\left(\dfrac{a \sin w}{1 - a\cos w}\right)}$

$\sqrt{(1 - a\cos w)^2 + (a \sin w)^2} = \sqrt{1 - 2a\cos w + a^2\cos^2 w + a^2 \sin^2 w}$

$= \sqrt{1 - 2a\cos w + a^2}$

f. magnitude is always 1



6. $H(z) = \dfrac{(z-e^{j\omega_0})(z-e^{-j\omega_0})}{z^2} = \dfrac{z^2 - ze^{j\omega_0} - ze^{-j\omega_0} + 1}{z^2} = \dfrac{z^2 - 2z\cos\omega_0 + 1}{z^2}$

$= 1 - 2z^{-1}\cos\omega_0 + z^{-2}$

a. $b = (1, -2\cos\omega_0, 1)$

b. $\dfrac{\omega f_s}{2\pi} = f_0 \rightarrow \dfrac{\omega(8000)}{2\pi} = 1000 \rightarrow \omega = \dfrac{\pi}{4}$ $\quad -2\cos\frac{\pi}{4} = -\frac{2\sqrt{2}}{2} = -\sqrt{2}$

$b = (1, -\sqrt{2}, 1)$

c. The notch is very deep but very narrow.

d. $\dfrac{\omega(11250)}{2\pi} = 1000$, $\omega = \dfrac{8\pi}{45}$

$b = (1, -1.696, 1)$

This notch isn't as deep nor narrow. Still works as a notch filter tho.

7. $H(z) = \dfrac{(z-e^{j\omega_0})(z-e^{-j\omega_0})}{(z-pe^{j\omega_0})(z-pe^{-j\omega_0})}$

a. $\dfrac{z^2 - 2z\cos\omega_0 + 1}{z^2 - 1.8z\cos\omega_0 + .81}$ $\quad b = (1, -2\cos\omega_0, 1)$ $\quad a = (1, -1.8\cos\omega_0, .81)$

b. $\dfrac{\omega(8000)}{2\pi} = 1000$ $\quad \omega = \dfrac{\pi}{4}$

$b = (1, -\sqrt{2}, 1)$

$a = (1, -.9\sqrt{2}, .81)$

c. Looks like:

d. This filter is much better than the FIR. It is narrower and almost as deep. It looks almost perfect. This must be b/c our poles are closer to our zeros.